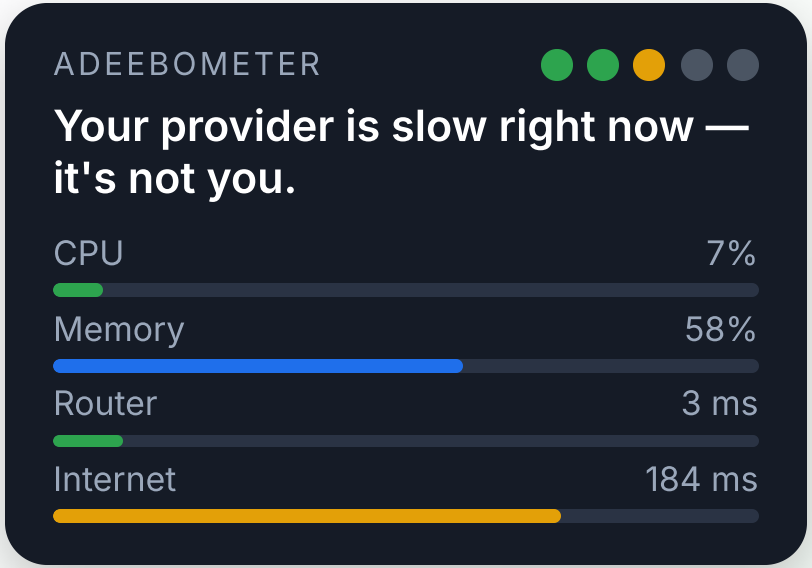
# AdeeboMeter

## Is it me or the network?

A small, quiet Windows companion. One glance tells you whether your PC, your Wi-Fi, your provider, DNS or the internet is the problem — and what to do about it.

**No cloud. No telemetry. Never listens.**
Every packet it sends is listed — and you can switch each one off.




The corner widget — faded, never takes a click

THE IDEA IN ONE BREATH

# Stop guessing. Get one sentence.

**Task Manager**
Is something eating my CPU?

**Command prompt**
`ipconfig`, `netsh wlan`…
what do these numbers mean?

**ping · tracert**
Some hops time out. Is that bad?

**Guess**
"Must be the Wi-Fi." Restart the router. Hope.

↓

**"Your Wi-Fi signal dropped to 38% when the calls started stuttering. Move closer to the access point."**

AdeeboMeter does all of that for you, quietly, every second — and puts the fix right after the finding.

IT LIVES IN TWO PLACES

# A glance, or the whole story.

Always on · penny-light

## The corner widget

- Faded to 30% (up to 100%), small or bigger; brightens when you hover
- Click-through — whatever is underneath keeps working
- Folds into a tiny chip when a window covers the corner
- CPU, memory, disk, network and a verdict in one sentence

On demand · tray icon or Ctrl+Alt+A

## The engineer view

- **Overview** · **Monitor** · **Investigate**
- **Network story** · **Time machine** · **Learn**
- Hover any **?** cursor for a plain-English explanation
- Esc, minimise or X = back to the corner — memory given back

DESIGN

# Five promises the whole design follows

### A penny app

A few dozen tiny jobs a minute. Anything heavy runs only while you look at it, and stops by itself.

### Never listens

No server, account or update check. It checks at every start that it opens no listening port.

### No admin rights

Only what Windows shows a normal user. If Windows says no, that part simply doesn't show.

### One readable file

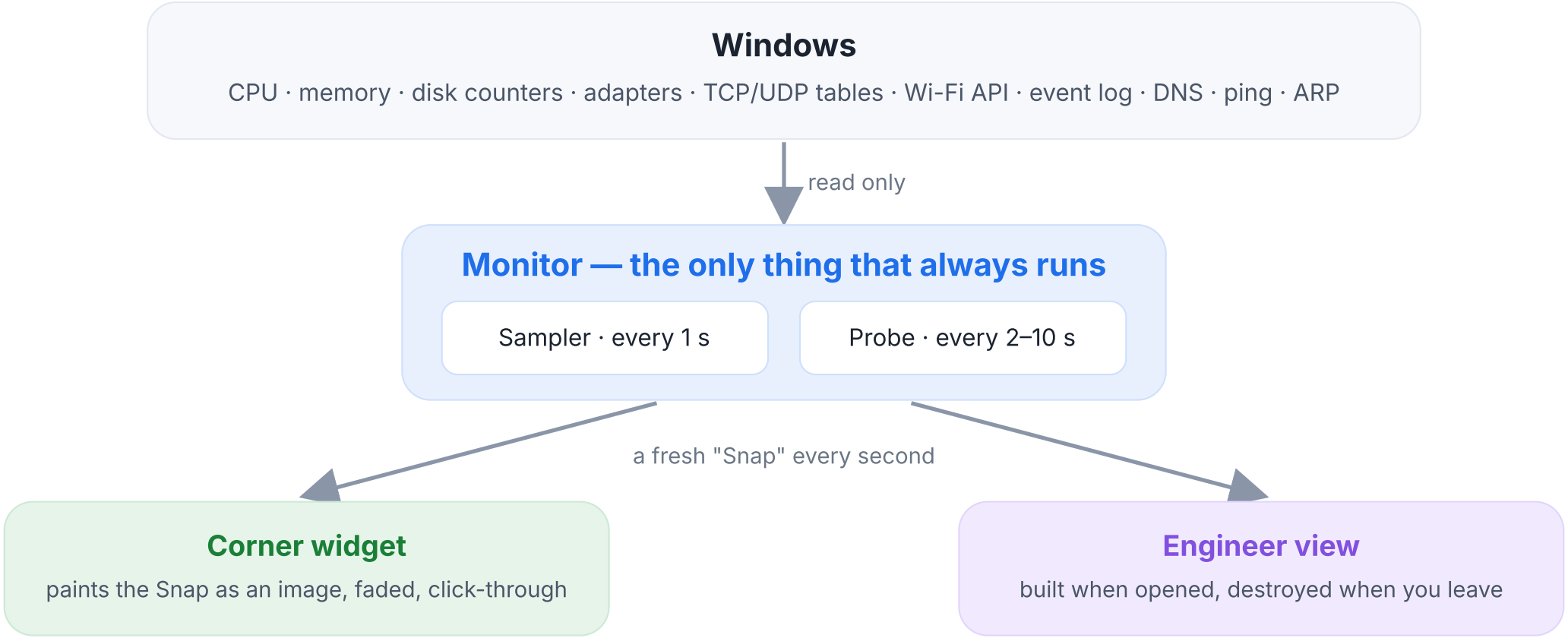
One C# file, built by the compiler already inside Windows. Nothing to install, nothing hidden.

### A companion

Every message has a friendly and a plain version. The fix always comes right after the joke.

THE BIG PICTURE

# One heart, two faces.




The screen only ever reads finished snapshots — painting can never slow the Monitor down.

THE LIFE OF ADEEBOMETER

# From start to exit, nothing left behind.

1

### Start

One copy only. Loads settings. Refuses to run if it is listening on any port.

2

### Run

The Monitor's two threads start. Tray icon and corner widget appear.

3

### Open

Tray click or Ctrl+Alt+A. The Monitor speeds up while you look.

4

### Back

Esc, minimise, X or *Back to corner*. The view closes and gives its memory back.

5

### Exit

Tray → Exit. Helper tools die with the app (a Windows job object); temporary captures are deleted.

THE HEART · MONITOR

# Two threads do all the watching.

## Sampler — every second

- CPU, memory, disk and network bytes
- TCP/IP health: retransmits, resets, failed connections, unreachables, UDP drops
- Every 10 s it **judges**: OK / Busy / Act, and one sentence
- Optional: one line to today's 7-day history file

## Probe — every 2 to 10 seconds

- Router ping · internet + DNS check (cache skipped)
- Adapters, VPN, DNS server, MTU — changes become events
- Gateway spoof watch: the router's MAC must not change
- Wi-Fi signal, roams, drops — and Windows' reason for a drop
- Apps and connections · Teams calls get the fast pace

HOW THE VERDICT IS FOUND

# The first dot that isn't green is where trouble starts.




### Everything after it is a victim, not a cause

If your provider is slow, DNS and the internet look slow too. AdeeboMeter blames the first broken link, not the last one.

### Cause hints line up *when* things happened

"TCP retransmits (12/s) started when Wi-Fi signal fell to 40%." Click a dot to jump to the page that explains it.

ENGINEER VIEW · TOUR

# Seven places. Everything you'd dig out of a dozen tools.

### Overview
The verdict, router, internet, DNS and traffic right now, what needs attention, today.

### Monitor
Connections (**Sniff** into Wireshark), your Wi-Fi or cable, what's running, live capture.

### Investigate
Pick the problem; it runs the checks. Check a destination, health check.

### Network story
Follow an app from your PC to the server, box by box, and time each step.

### Time machine
Replay any minute, the week as a heatmap, data usage, calls, new devices.

### Learn
46 topics, two lines and a picture each — from "the layers" to svchost.

### Toolbox
56 tools for the path, protocols, Windows and the PC. Type to find one.

### Hover the ? cursor
219 terms and 72 programs explained. Ctrl+K jumps to anything.

INVESTIGATE

# Start from the problem, not the protocol.

## Six questions people actually ask

- The internet is slow
- A website or app won't open
- Calls freeze or sound robotic
- I keep disconnecting
- It works at home, not on VPN or at work
- My PC is slow

Each runs the checks AdeeboMeter already has and answers in the same four parts: **Finding · What I observed · Not established · Next test**.

## Every fact says where it came from

- **measured** AdeeboMeter timed it itself
- **Windows-reported** Windows said so
- **seen in packets** Wireshark decoded it
- **inferred** a conclusion from the others
- **unavailable** this PC doesn't let us see it

**Details** (top right) adds raw values and the second each one was read.

TOOLBOX

# 56 tools. Type what you need.

### Path & routing

MTU per hop, MTR, the way back, which route, your gateway (VRRP / HSRP), routing protocols on your segment, IPv6.

### Is it reachable?

Port test, TCP ping, HTTP timing, proxy, Teams UDP, SSH, mail with STARTTLS, RADIUS, TACACS+.

### Speed & quality

iperf3 client to your server, sustained ping, TWAMP-light to your reflector, nearest cloud.

### Names & time

dig and the delegation from the root, encrypted DNS, hosts file, VPN name rules, NTP.

### Network devices

Neighbours (ARP), SNMP, syslog, your own router (CGNAT, hairpin), ping sweep, Wake-on-LAN, subnets.

### Windows

802.1X, domain controllers, Group Policy, expiring certificates, firewall rules, what's listening, the event log.

### The PC

Memory pressure, disk, heat, frozen windows, leaks, crashes, battery, where the space went.

### Always the client

Runs only when you press Run. Never listens, never logs in, never sends a routing packet.

NETWORK STORY · LEARN

# Follow one app from your PC to the server.

**Teams** → Windows (socket) → Wi-Fi → Router → Provider → Internet → **Microsoft**

### Ready when you open it

The busiest app is picked for you; nothing to type. Click a box for what happens there.

### Where the time goes

Name · connect · lock · first byte, as one bar. Real packets with Wireshark.

### Learn in a glance

Two lines, a picture, and one "if it goes wrong" line. People who want the textbook read it elsewhere.

YOUR WI-FI · YOUR CABLE

# What your radio and your cable are really doing.

### Radio

Exact access point, 2.4 / 5 / 6 GHz, channel, **width** (20–160 MHz), Wi-Fi generation, signal in % and dBm, **roaming help** (802.11k/v/r) — read from the access point's own beacons.

### Who else is on the air

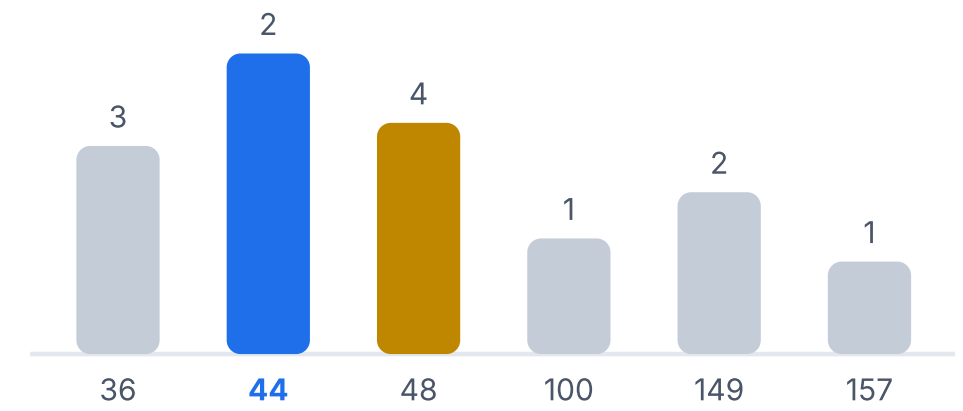
How many access points share or overlap your channel, and how busy the air is.

### Windows' diary

Every Wi-Fi drop in 24 h with the reason Windows recorded; DHCP trouble.

### Cable

Negotiated speed (100 Mb/s = bad-cable sign) and **Find my switch port**: switch, port, VLAN from LLDP/CDP. Nothing is sent.



WHAT'S RUNNING

# Every program, with a story — not just a name.

Click any program for five answers:

- **What it does**
- **How it works**
- **Why you need it**
- **What happens without it**
- **Can I close it?**

Programs not in the guide are described from the file itself — description, publisher, folder. AdeeboMeter **only explains; it never closes anything.**

**Spotify** · your app — safe to close

**OneDrive** · syncs your files — leave it

**svchost** · Windows services — Windows needs it

**VPN client** · company tool — managed by IT

CHECK A DESTINATION

# Eight steps, from the name to the last hop.

1
### DNS
Real lookups, cache skipped. A second opinion from 1.1.1.1 catches filters.

2
### Route
Which adapter Windows would use — found without sending anything.

3
### Ping
Four echoes: delay and loss.

4
### TCP
The three-way handshake to your port, timed.

5
### TLS
1.2 / 1.3 only, Windows validates. Untrusted? Nothing is sent.

6
### HTTP
One HEAD request: status line and time to first byte.

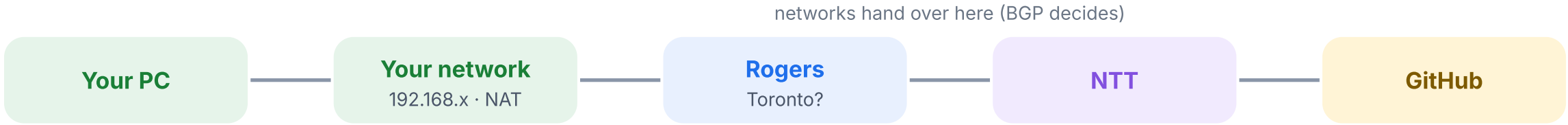
7
### Path MTU
"Don't fragment" pings find the biggest packet that fits.

8
### Path
MTR-style trace: 3 rounds per hop, all hops at once.

If a check fails, a **sign-in page check** catches hotel and airport portals. Pin up to 5 destinations and **Run all**.

THE PATH STORY

# Who carries your traffic — and what's in the way.




### NAT

Double NAT and carrier-grade NAT (100.64.x.x) spotted from the hop addresses.

### Tunnels

From the packet size: 1492 = PPPoE; smaller = VPN, LTE or satellite.

### Firewalls

A refusal faster than the server is far = something in between said no. Ping works, port silent = dropped.

### And more

Load-balanced hops, reroutes, where the delay jumps, hidden tunnel hops, the return path — and, with online lookups, the network that owns each hop.

LIVE CAPTURE · TIME MACHINE

# Deep when you ask. Silent when you don't.

Only while you press Start · stops after 15 min

## Live capture (Wireshark's engine)

- What your traffic is made of; top conversations named from the traffic itself
- Wireshark's expert warnings; bytes per app
- **Call quality:** RTP loss, jitter (RFC 3550), MOS score (ITU-T E-model) — audio stays encrypted
- **Rogue DHCP:** two servers answering on one network

History stays on this PC · 3 days by default (1–90)

## Time machine

- **Replay:** drag to any minute — the dots, the numbers, what changed around then
- **The week:** 7 × 24 squares; click one to replay that hour
- Data usage per network, "slower than usual", after-call summaries, new devices
- One line a minute, older days compressed, oldest deleted every night

PRIVACY

# What leaves your PC — and what never does.

| WHAT | WHEN | WHERE TO |
|---|---|---|
| **Router ping** | every 2–10 s (Settings) | your own router |
| **Internet + DNS check** | every 15–60 s, or off | 1.1.1.1 / 8.8.8.8 and your DNS server |
| **Provider name** | once per network | a 6-hop trace toward 1.1.1.1, one reverse-DNS lookup |
| **Checks you start** | when you press Check, Investigate or Time each step | where you asked, plus one DNS question to 1.1.1.1 |
| **DNS speed test · pinned watch** | when you press it · opt-in, every 30 s | 1.1.1.1, 8.8.8.8, 9.9.9.9 · the destination you pinned |
| **Sign-in page check** | only if a check fails (can be off) | msftconnecttest.com — the page Windows itself uses |
| **Toolbox** | only when you press Run | the address you typed, or the one the tool names on screen |
| **Online lookups** | opt-in, 5 min–3 h, then off by themselves | public IP addresses only → RIPEstat, PeeringDB |
| **Wi-Fi details · live capture** | — | nothing leaves; Windows is asked locally |

**Never:** cloud · account · telemetry · an updater · ads · crash reports sent · listening ports

SECURITY

# Hardened by design, proven by checks.

### No shell, ever

Windows tools by full System32 path, fixed arguments, a timeout, inside a job object that dies with the app.

### Inputs validated

Destinations checked strictly before anything runs. Capture paths fenced inside AdeeboMeter's own folder.

### TLS done right

Only TLS 1.2 / 1.3; Windows validates certificates; nothing is sent over an untrusted connection.

### DLLs only from System32

For the whole process — a DLL planted next to the exe in Downloads is never loaded.

### Nothing to hijack

No updater, no binary serializers, no XML parser, no secrets in the code. A RADIUS secret is masked and never kept.

### A provable build

Built by GitHub from this source, actions pinned to exact commits; SHA-256 and build provenance on every file. 0 analyzer findings.

STAYING LIGHT

# A penny in the background.

**~0.5%**
of one CPU core with only the widget

**2–3%**
with the engineer view open — zero the moment you leave

**64 MB**
memory budget, shown live in the footer

**18·36·74**
checks a minute: Penny · Balanced · Engineer

### Memory stays flat

Fixed-size rings; capped lists (300 events, 3,000 conversations, 7 days).

### Paints only on change

The widget checks the mouse 20×/s near the corner, 4×/s otherwise, 2×/s when hidden.

### Nothing leaks

Windows, timers, bitmaps and helper tools released on close — and the self-test proves it.

PROOF, NOT PROMISES

# AdeeboMeter.exe --selftest

30 checks on your own PC, written to `AdeeboMeter-selftest.txt`:

- Counters, socket tables and DNS reading work
- **Never listens**
- Bad destinations are rejected
- The engineer view opens in under 1 second
- **No handle leaks** over 30 open/close cycles
- Every label has an explanation
- Parsers give the right answers on known samples: Wi-Fi, LLDP/CDP, event log, RTP, the path story
- Memory stays inside the 64 MB budget
- `--deeptest`: hostile input into every tool, repeats to catch leaks, the UI under stress

```
AdeeboMeter 1.1.1 self-test
PASS never listens
PASS socket tables
PASS destination validation
PASS engineer view opens < 1 s
PASS 30 open/close, no leaks
PASS Wi-Fi information elements
PASS LLDP / CDP parser
PASS RTP call quality
…
ALL PASSED
```

HONEST LIMITS

# What Windows doesn't let us do without admin or the cloud.

### AS numbers / BGP data

Need an internet service — only through the opt-in, timed online lookups. Otherwise router names tell the story.

### TCP / UDP traceroute

Needs raw sockets (admin) — traces use ICMP.

### Per-connection RTT

Needs admin to switch on — per-app traffic comes from live capture instead.

### The DNS cache

Admin-only on some Windows builds — then names come from live capture.

### Speed tests

Only to a server you run (iperf3, TWAMP-light) — AdeeboMeter never listens.

### Inside a tunnel or your ISP

Can only be inferred, never seen.

GET IT

# One file. It installs itself.

1

### Download

The zip from the website or GitHub Releases. Open it.

2

### Double-click AdeeboMeter.exe

Choose **Install**. It goes in just for you — a Start menu entry, an entry in Settings › Apps, no admin rights.

3

### Look at the corner

It's already running, bottom-right. Tray icon or Ctrl+Alt+A for the full story.

### Update

Run the new exe, choose **Update**. Your settings come along; the old version steps aside.

### Remove

Settings › Apps, or the tray menu. Everything goes — folder, settings, history.

### Or don't install

*Just run it from here* keeps it portable. Not code-signed: SmartScreen says *More info → Run anyway*.

# Is it me or the network?

Now you know — in one glance.

No cloud. No telemetry. Never listens.
Every packet it sends is listed — and you can switch each one off.

`adeeb111.github.io/AdeeboMeter`